

Sulfur is the 10th most abundant element on Earth and is an essential element in biological systems. Sulfur can form a variety of chemical compounds and will react with nearly every element except Au, Pt, Ir, Te, and the noble gases. A list of several common inorganic compounds of sulfur is given in Table 1.

Table 1 Selected Common Inorganic Sulfur Compounds

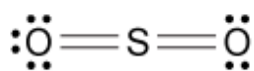
Compound	Molar mass (g/mol)
H ₂ S	32.08
SO ₂	64.07
SO ₃	80.07
H ₂ SO ₄	98.08
Na ₂ SO ₄	142.04
K ₂ SO ₄	174.26
SF ₆	146.06
S ₂ Cl ₂	135.04
S ₄ N ₄	184.29

Reduction of elemental sulfur by bacteria during anaerobic digestion produces the toxic, foul-smelling gas H₂S. Oxidation of elemental sulfur via combustion or volcanic processes yields gases of SO₂ and SO₃. When dissolved in water, both SO₂ and SO₃ produce acidic solutions, and SO₃ reacts with water to form the acid H₂SO₄. Neutralization of H₂SO₄ can provide a variety of salts, including Na₂SO₄.

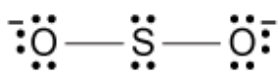
Elemental sulfur can also react with halogen gases to form sulfur halides such as the industrially important insulator gas SF₆ and the synthetic reagent S₂Cl₂. S₂Cl₂ facilitates the formation of other sulfur compounds. For example, reacting S₂Cl₂ with NH₃ produces the sulfur pnictide S₄N₄.

Which of the compound diagrams shown below depicts a valid Lewis structure for sulfur dioxide with the correct number of valence electrons and

the correct net charge?



Compound 1



Compound 2

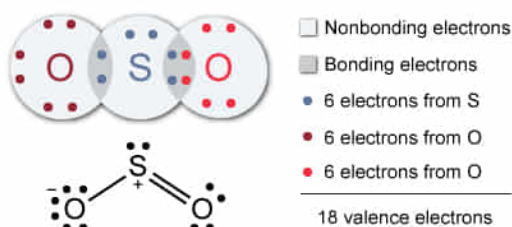
- ☐ A. Compound 1 only [17%]
☐ B. Compound 2 only [39%]
☐ C. Both compounds [6%]
✔ ☒ D. Neither compound [36%]

Correct

36% answered correctly

Explanation:

Lewis structure of sulfur dioxide following the octet rule



When bonding involves only *s* and *p* orbitals and follows the octet rule, each atom has only 8 valence electrons.

©UWorld

Lewis structures symbolically depict chemical compounds by using chemical symbols to represent atoms and dots placed around the symbols to represent the **valence electrons** and their **bonding configuration**. Dots placed between two symbols represent shared electrons involved in bonding whereas dots placed elsewhere around a symbol represent unshared, nonbonding electrons. Each shared pair of electrons forms one bond (often shown as a line).

When **drawing Lewis structures**, dots are placed around the elemental symbols in a way that gives each atom a **full valence shell** of electrons. Atoms that follow the **octet rule** use only *s* and *p* orbitals and achieve a full valence shell by having eight valence electrons (an octet). **Exceptions** to the octet rule occur for small atoms that have only an *s* orbital (H and He) and for larger atoms in Period 3 and beyond when *d* orbitals are utilized.

To draw the Lewis structure of SO_2 , the two oxygen atoms can be initially joined by a single bond with S as the central (least electronegative) atom, but to achieve an octet around each atom, an additional pair of nonbonding electrons must be moved to form a double bond (two shared pairs of electrons).

(Choice A) The diagram for Compound 1 is not valid because it has 10 valence electrons around one oxygen atom. Atoms in Period 2 (eg, oxygen) form bonds using only *s* and *p* orbitals. As such, these atoms follow the octet rule and cannot have a bonding configuration involving more than 8 valence electrons.

(Choice B) The diagram for Compound 2 is not valid because it has a total of 20 valence electrons and has a net charge of -2 . SO_2 is a neutral compound and has no net charge. Atoms from Group 16 each contribute only 6 valence electrons. Therefore, a diagram for SO_2 should have a total of only $3 \times 6 = 18$ valence electrons (not 20 valence electrons).

(Choice C) Compound 1 does not follow the octet rule, and Compound 2 has too many valence electrons and an incorrect net charge for SO_2 .

Educational objective:

Lewis structures symbolically depict chemical compounds by using chemical symbols to represent atoms and dots placed around the symbols to represent the valence electrons and their bonding configuration. Dots are arranged to give each atom a full valence shell of electrons following either the octet rule or a valid exception.

Time spent:0 sec

Subject:
General Chemistry

QID:402495

System:
Interactions of Chemical Substances

Sulfur is the 10th most abundant element on Earth and is an essential element in biological systems. Sulfur can form a variety of chemical compounds and will react with nearly every element except Au, Pt, Ir, Te, and the noble gases. A list of several common inorganic compounds of sulfur is given in Table 1.

Table 1 Selected Common Inorganic Sulfur Compounds

Compound	Molar mass (g/mol)
H ₂ S	32.08
SO ₂	64.07
SO ₃	80.07
H ₂ SO ₄	98.08
Na ₂ SO ₄	142.04
K ₂ SO ₄	174.26
SF ₆	146.06
S ₂ Cl ₂	135.04
S ₄ N ₄	184.29

Reduction of elemental sulfur by bacteria during anaerobic digestion produces the toxic, foul-smelling gas H₂S. Oxidation of elemental sulfur via combustion or volcanic processes yields gases of SO₂ and SO₃. When dissolved in water, both SO₂ and SO₃ produce acidic solutions, and SO₃ reacts with water to form the acid H₂SO₄. Neutralization of H₂SO₄ can provide a variety of salts, including Na₂SO₄.

Elemental sulfur can also react with halogen gases to form sulfur halides such as the industrially important insulator gas SF₆ and the synthetic reagent S₂Cl₂. S₂Cl₂ facilitates the formation of other sulfur compounds. For example, reacting S₂Cl₂ with NH₃ produces the sulfur pnictide S₄N₄.

The chemical structures of sulfur compounds H₂S, SF₆, S₂Cl₂, and S₄N₄ each contain only single (sigma) bonds. Based on trends in the atomic radii of the

elements, the compound that contains the longest bond between sulfur and an atom of another element is:

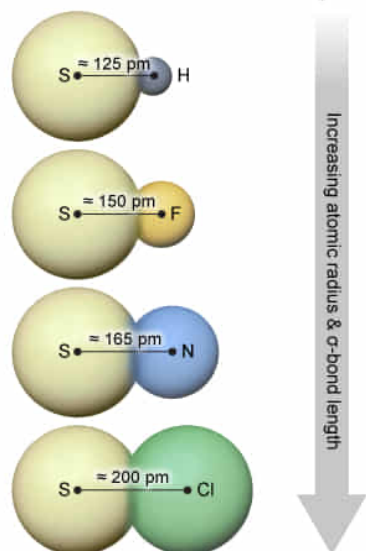
- ☐ A. H_2S [32%]
- ☐ B. SF_6 [8%]
- ✓ ☒ C. S_2Cl_2 [46%]
- ☐ D. S_4N_4 [12%]

Correct

47% answered correctly

Explanation:

Atomic radius & σ -bond length



The length of the σ -bond increases as the sum of the atomic radii increases.

©UWorld

Covalent bonds between atoms are formed by **sharing electrons**, which is made possible through the overlap of the **atomic orbitals** from each atom. Covalent bonds made by the **end-to-end overlap** of atomic orbitals are called **sigma (σ) bonds**, whereas covalent bonds made by the side-to-side overlap of p orbitals are called **pi (π) bonds**. A single covalent bond consists only of a σ bond, a double bond consists of one σ bond and one π bond, and a triple bond consists of one σ bond and *two* π bonds.

Because a σ bond involves end-to-end overlap of two orbitals, the **length of a σ bond** can be estimated as the **sum of the atomic radii** of the bonded atoms. Bonding between atoms with larger atomic radii positions the atomic nuclei farther apart and results in a longer σ bond (and vice versa).

On the periodic table, **atomic radii** tend to **decrease across a row** due to electrons being pulled closer by an increasing **effective nuclear charge**, whereas the atomic radii **increase down a column** as the number of electron shells increases. The changes in the radii across a row are smaller than the changes down a column because adding additional electron shells has a much greater effect on the atomic radius.

In this question, the given compounds all have σ bonds to a sulfur atom. As a result, the longest bond between sulfur and another element will be formed with the element that has the largest atomic radius. Because Cl has a larger atomic radius than H, N, or F, the S–Cl bond in S_2Cl_2 will be the longest bond with sulfur among the given compounds.

(Choices A, B, and D) Atoms of H, N, and F each have a smaller atomic radius than Cl because Cl is in Period 3 and has an additional electron shell farther from the nucleus. Therefore, the S–H bonds in H_2S , the S–F bonds in SF_6 , and the S–N bonds in S_4N_4 are all *shorter* than the S–Cl bonds in S_2Cl_2 .

Educational objective:

Covalent sigma bonds are made by sharing electrons through the end-to-end overlap of atomic orbitals. The length of a sigma bond can be estimated as the sum of the atomic radii of the bonded atoms. Atomic radii tend to decrease across a row and increase down a column on the periodic table.

Time spent:0 sec

Subject:
General Chemistry

QID:402496

System:
Interactions of Chemical Substances

Sulfur is the 10th most abundant element on Earth and is an essential element in biological systems. Sulfur can form a variety of chemical compounds and will react with nearly every element except Au, Pt, Ir, Te, and the noble gases. A list of several common inorganic compounds of sulfur is given in Table 1.

Table 1 Selected Common Inorganic Sulfur Compounds

Compound	Molar mass (g/mol)
H ₂ S	32.08
SO ₂	64.07
SO ₃	80.07
H ₂ SO ₄	98.08
Na ₂ SO ₄	142.04
K ₂ SO ₄	174.26
SF ₆	146.06
S ₂ Cl ₂	135.04
S ₄ N ₄	184.29

Reduction of elemental sulfur by bacteria during anaerobic digestion produces the toxic, foul-smelling gas H₂S. Oxidation of elemental sulfur via combustion or volcanic processes yields gases of SO₂ and SO₃. When dissolved in water, both SO₂ and SO₃ produce acidic solutions, and SO₃ reacts with water to form the acid H₂SO₄. Neutralization of H₂SO₄ can provide a variety of salts, including Na₂SO₄.

Elemental sulfur can also react with halogen gases to form sulfur halides such as the industrially important insulator gas SF₆ and the synthetic reagent S₂Cl₂. S₂Cl₂ facilitates the formation of other sulfur compounds. For example, reacting S₂Cl₂ with NH₃ produces the sulfur pnictide S₄N₄.

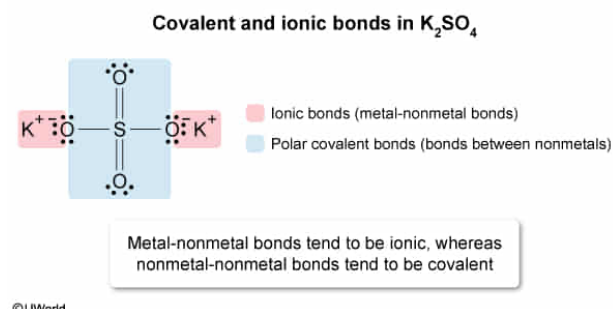
Which of the following sulfur compounds from Table 1 has a chemical structure that contains both ionic and covalent bonds?

- ☐ A. Hydrogen sulfide [1%]
- ☐ B. Sulfuric acid [10%]
- ☐ C. Sulfur trioxide [1%]
- ☒ D. Potassium sulfate [86%]

Correct

86% answered correctly

Explanation:



Ionic bonds are formed between two atoms when the valence electrons from one atom are transferred to another atom, resulting in the formation of charged ions held together by strong electrostatic attractions. In contrast, **covalent bonds** are made between two atoms by sharing electrons through the end-to-end overlap of atomic orbitals.

The type of bond that forms between two atoms depends on the **relative difference** in **electronegativity** between the atoms. A large difference in electronegativity promotes ionic bond formation, but a small difference in electronegativity promotes covalent bond formation. As a result, ionic bonds generally form between a **metal and a nonmetal** whereas covalent bonds generally form between **two nonmetals**; however, some exceptions do exist because the actual ionic character of a bond exists on a **continuum**.

Using the metal and nonmetal classifications of the elements as generalization for determining bonding types in the four compounds given in this question, only potassium sulfate (K_2SO_4) contains a metal. Therefore, potassium must form a metal-nonmetal (ionic) bond in the structure in addition to the nonmetal-nonmetal (covalent) bonds present between the nonmetal atoms in the structure.

(Choices A, B, and C) Hydrogen sulfide (H_2S), sulfuric acid (H_2SO_4), and sulfur trioxide (SO_3) each contain only nonmetals in their chemical structures. Although the electronegativity differences between some of the atoms in these compounds produce **polar covalent bonds** with a **dipole**, the differences are not large enough for these bonds to be considered ionic.

Although the O-H bonds of H_2SO_4 do readily break to give H^+ and SO_4^{2-} in water, this does not indicate that the bonds are ionic. Rather, these bonds readily ionize because they are weak, and ionization is favorable because the resulting charges formed in SO_4^{2-} are stabilized by **resonance**.

Educational objective:

The type of bond formed between two atoms depends on the relative difference in electronegativity between the atoms. Atoms with a large difference in electronegativity (usually a metal and a nonmetal) form ionic bonds. Atoms with a small difference in electronegativity (usually two nonmetals) form covalent bonds.

Time spent:0 sec**Subject:**
General Chemistry**QID:**402497**System:**
Interactions of Chemical Substances

Sulfur is the 10th most abundant element on Earth and is an essential element in biological systems. Sulfur can form a variety of chemical compounds and will react with nearly every element except Au, Pt, Ir, Te, and the noble gases. A list of several common inorganic compounds of sulfur is given in Table 1.

Table 1 Selected Common Inorganic Sulfur Compounds

Compound	Molar mass (g/mol)
H ₂ S	32.08
SO ₂	64.07
SO ₃	80.07
H ₂ SO ₄	98.08
Na ₂ SO ₄	142.04
K ₂ SO ₄	174.26
SF ₆	146.06
S ₂ Cl ₂	135.04
S ₄ N ₄	184.29

Reduction of elemental sulfur by bacteria during anaerobic digestion produces the toxic, foul-smelling gas H₂S. Oxidation of elemental sulfur via combustion or volcanic processes yields gases of SO₂ and SO₃. When dissolved in water, both SO₂ and SO₃ produce acidic solutions, and SO₃ reacts with water to form the acid H₂SO₄. Neutralization of H₂SO₄ can provide a variety of salts, including Na₂SO₄.

Elemental sulfur can also react with halogen gases to form sulfur halides such as the industrially important insulator gas SF₆ and the synthetic reagent S₂Cl₂. S₂Cl₂ facilitates the formation of other sulfur compounds. For example, reacting S₂Cl₂ with NH₃ produces the sulfur pnictide S₄N₄.

The anion SO₃²⁻ and sulfur trioxide, SO₃, have the same elemental composition. These two chemical species are:

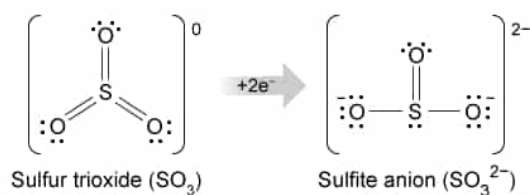
- ☐ A. the same compound, because both have the same elemental composition. [21%]
- ☐ B. the same compound, because both have the same number of electrons and bonding configuration. [2%]
- ☐ C. different compounds, because the atoms in SO₃²⁻ are held together by ionic bonds. [1%]
- ☒ D. different compounds, because SO₃²⁻ has two more electrons and a different bonding configuration. [75%]

Correct

75% answered correctly

Explanation:

Lewis Structures of SO_3 & SO_3^{2-}



SO_3 and SO_3^{2-} are different compounds with a different number of electrons and a different bonding configuration.

©UWorld

Polyatomic ions are covalently bonded groups of two or more atoms that have a net **formal charge**. If the total number of electrons exceeds the total number of protons in the group, the polyatomic ion will be negatively charged (and vice versa). Each electron present in a polyatomic group provides one unit of negative charge. For example, in SO_3^{2-} (a sulfite anion), the number of protons in the group is exceeded by two electrons.

Some polyatomic ions have an **elemental composition** that is the same as a neutral (uncharged) covalent molecule, but the polyatomic ions and the neutral covalent compounds are chemically distinct. For example, the sulfite anion (SO_3^{2-}) and the covalent compound sulfur trioxide (SO_3) both consist of one S atom and three O atoms, but the two species are not the same compound. As seen by comparing the Lewis structures for each, both the **number of electrons** and the **bonding configuration** of SO_3^{2-} are different from those of SO_3 .

The additional pair of nonbonding electrons on the S atom of SO_3^{2-} causes the structure of SO_3^{2-} to adopt a **trigonal pyramidal geometry** with a net charge of -2 . In contrast, SO_3 lacks the nonbonding electrons on the S atom, which causes the molecule to adopt a **trigonal planar geometry** with no net charge. These characteristics give SO_3 and SO_3^{2-} different chemical and physical properties.

(Choice A) The same set of atoms can bond in different ways. For two chemical species to be the same, both the elemental composition *and* the electronic bonding configuration must be the same.

(Choice B) SO_3^{2-} has two more electrons than SO_3 , and SO_3 has a different bonding configuration than SO_3^{2-} .

(Choice C) Although SO_3^{2-} (as a structural unit) is a polyatomic ion, the S and O atoms within SO_3^{2-} are held together by *covalent* bonds.

Educational objective:

Identical chemical species must have the same elemental composition, the same number of electrons, and the same orbital bonding configuration. The net charge of a structure is equal to the sum of all charges present.

Time spent:0 sec

Subject:
General Chemistry

QID:402498

System:
Interactions of Chemical Substances

Sulfur is the 10th most abundant element on Earth and is an essential element in biological systems. Sulfur can form a variety of chemical compounds and will react with nearly every element except Au, Pt, Ir, Te, and the noble gases. A list of several common inorganic compounds of sulfur is given in Table 1.

Table 1 Selected Common Inorganic Sulfur Compounds

Compound	Molar mass (g/mol)
H ₂ S	32.08
SO ₂	64.07
SO ₃	80.07
H ₂ SO ₄	98.08
Na ₂ SO ₄	142.04
K ₂ SO ₄	174.26
SF ₆	146.06
S ₂ Cl ₂	135.04
S ₄ N ₄	184.29

Reduction of elemental sulfur by bacteria during anaerobic digestion produces the toxic, foul-smelling gas H₂S. Oxidation of elemental sulfur via combustion or volcanic processes yields gases of SO₂ and SO₃. When dissolved in water, both SO₂ and SO₃ produce acidic solutions, and SO₃ reacts with water to form the acid H₂SO₄. Neutralization of H₂SO₄ can provide a variety of salts, including Na₂SO₄.

Elemental sulfur can also react with halogen gases to form sulfur halides such as the industrially important insulator gas SF₆ and the synthetic reagent S₂Cl₂. S₂Cl₂ facilitates the formation of other sulfur compounds. For example, reacting S₂Cl₂ with NH₃ produces the sulfur pnictide S₄N₄.

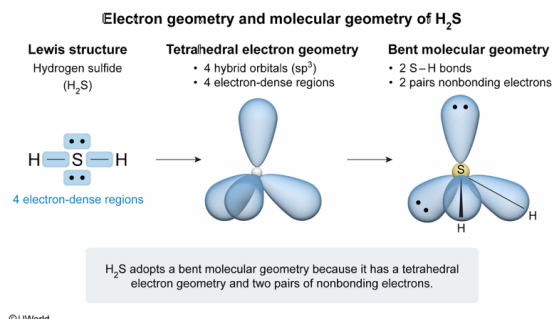
What is the molecular geometry of hydrogen sulfide?

- ✔ ☒ A. Bent [84%]
☐ B. Trigonal planar [4%]
☐ C. Tetrahedral [6%]
☐ D. Trigonal pyramidal [4%]

Correct

84% answered correctly

Explanation:



Hybridization is the process by which two or more **atomic orbitals** (*s*, *p*, *d*, *f*) combine to form new **hybrid orbitals** that have different **shapes** and spatial orientations than the atomic orbitals that produce them. Conservation of orbitals dictates that the number of hybrid orbitals formed must equal the number of atomic orbitals that hybridize. For example, in **sp³ hybridization**, 4 atomic orbitals hybridize, so 4 hybrid orbitals are formed.

When hybrid orbitals are formed, the orbitals will adopt a spatial orientation that maximizes separation between the orbitals to minimize the repulsion between the electron-dense regions. This action forms an **electron geometry** for the **electron-dense regions** that acts as a frame on which bonding may occur. However, the specific **molecular geometry** (shape) of a compound depends on the number and orientation of the **bonds** around the **central atom**. As a result, two compounds with the same electron geometry may have different molecular geometries depending on the number of pairs of **nonbonding electrons** present in each.

The Lewis structure of hydrogen sulfide (H₂S) shows that the central sulfur atom has two pairs of nonbonding electrons and two bonds (a total of four electron-dense areas). The four electron-dense areas of the S atoms give H₂S a tetrahedral electron geometry, but because two of the four electron-dense regions are nonbonding electrons, the molecular geometry (the shape made by the bonded atoms) is **bent**.

(Choice B) A trigonal planar molecular geometry results from a trigonal planar electron geometry without any nonbonding electrons. H₂S has a *tetrahedral* electron geometry.

(Choice C) A tetrahedral molecular geometry results from a tetrahedral electron geometry that lacks any nonbonding electrons. H₂S has *two pairs* of nonbonding electrons.

(Choice D) A trigonal pyramidal molecular geometry results from a tetrahedral electron geometry with only one pair of nonbonding electrons.

Educational objective:

Atomic orbitals can combine into hybrid orbitals, which adopt an electron geometry that maximizes the separation and minimizes the repulsion between the electron-dense orbital regions. The molecular geometry (shape) of a molecule depends on the electron geometry and number of nonbonding electron pairs of the central atom.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402499

System:
Interactions of Chemical Substances

Sulfur is the 10th most abundant element on Earth and is an essential element in biological systems. Sulfur can form a variety of chemical compounds and will react with nearly every element except Au, Pt, Ir, Te, and the noble gases. A list of several common inorganic compounds of sulfur is given in Table 1.

Table 1 Selected Common Inorganic Sulfur Compounds

Compound	Molar mass (g/mol)
H ₂ S	32.08
SO ₂	64.07
SO ₃	80.07
H ₂ SO ₄	98.08
Na ₂ SO ₄	142.04
K ₂ SO ₄	174.26
SF ₆	146.06
S ₂ Cl ₂	135.04
S ₄ N ₄	184.29

Reduction of elemental sulfur by bacteria during anaerobic digestion produces the toxic, foul-smelling gas H₂S. Oxidation of elemental sulfur via combustion or volcanic processes yields gases of SO₂ and SO₃. When dissolved in water, both SO₂ and SO₃ produce acidic solutions, and SO₃ reacts with water to form the acid H₂SO₄. Neutralization of H₂SO₄ can provide a variety of salts, including Na₂SO₄.

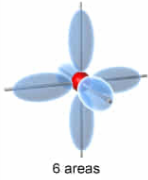
Elemental sulfur can also react with halogen gases to form sulfur halides such as the industrially important insulator gas SF₆ and the synthetic reagent S₂Cl₂. S₂Cl₂ facilitates the formation of other sulfur compounds. For example, reacting S₂Cl₂ with NH₃ produces the sulfur pnictide S₄N₄.

Which of the following compounds from Table 1 has a sulfur atom with a d^2sp^3 hybridization state?

- ☐ A. SO₂ [2%]
- ✓ ☒ B. SF₆ [86%]
- ☐ C. SO₃ [5%]
- ☐ D. S₂Cl₂ [6%]

Correct
86% answered correctly

Explanation:

Hybridization and geometry of SF ₆			
Hybridization	Electron-dense areas	Electron geometry	Molecular geometry
d^2sp^3	 6 areas	Octahedral	 SF ₆ (octahedral)

During **hybridization**, two or more **atomic orbitals** (s , p , d , f) combine to make **hybrid orbitals** that adopt a spatial orientation with an **electron geometry** that minimizes the repulsion between the electron-dense regions. Due to conservation of orbitals, the number of hybrid orbitals that form must equal the number of atomic orbitals that hybridize.

Each hybrid orbital that is formed by an atom can either bond with another atom or can hold a nonbonding pair of electrons. As a result, the sum of the number of **bonding regions** and the number of **nonbonding electron pairs** around the central atom in a compound must be equal to the number of hybrid orbitals employed by the central atom.

A sulfur atom with a d^2sp^3 hybridization combines one s orbital, three p orbitals, and two d orbitals (six orbitals in total) to make six d^2sp^3 hybrid orbitals that provide six electron-dense regions that can either form six bonds or exist as a combination of bonding regions and nonbonding electron pairs that total to six.

Although S is the central atom in the structure of each compound given in this question, only SF₆ has six electron-dense regions (in the form of six S–F bonds) around the sulfur atom.

Therefore, only SF₆ has a d^2sp^3 hybridized S atom.

(Choice A) The S atom in SO₂ is surrounded by three electron-dense regions (one S=O bond, one S–O bond, and one pair of nonbonding electrons) and has sp^2 hybridization.

(Choice C) SO₃ has a S atom surrounded by three electron-dense regions (three S=O bonds and no nonbonding electron pairs) and has sp^2 hybridization.

(Choice D) In S₂Cl₂, each S atom is surrounded by four electron-dense regions (one S–S bond, one S–Cl bond, and two nonbonding electron pairs) and has sp^3 hybridization.

Educational objective:

In hybridization, the number of hybrid orbitals that form must be equal to the number of atomic orbitals that hybridize. The sum of the number of bonding regions and the number of nonbonding electron pairs around the central atom in a compound must be equal to the number of hybrid orbitals employed by the central atom.

Time spent: 0 sec

Subject:
General Chemistry

QID: 402500

System:
Interactions of Chemical Substances